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A Taxonomy of Critical AI System Characteristics for Use in Proxy System Testing

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***Abstract*— Safety and trust are two of the most important features in a critical system. A critical system is one that must be highly reliable in that it not only completes its mission but causes zero harm to the public. The problem is testing a critical system, especially if it employs artificial intelligence (AI). The challenge is critical AI systems (CAIS) may cause unpredictable events and conditions that cannot be modeled during critical error testing. Proxy systems (non-critical prototype) are needed to test the critical system. We present a five-dimensional CAIS taxonomy and a weighting system to map system characteristics to a testing proxy in order to determine equivalent proxy systems to build and test. Ultimately this CAIS taxonomy and weighting system is a way forward to develop a set of proxy systems to use for critical error testing.**

Index Terms—artificial intelligence, autonomous systems, AI taxonomy, critical systems.

I. Introduction

Critical systems are those in which failure can harm the public. When the critical system employs some form of artificial intelligence (AI), we call this critical AI systems (CAIS). Assuring the public that CAIS are safe, and that the outcomes are fair, is essential to achieving the intended purpose and to widespread adoption of CAIS. But providing such assurance is very challenging, particularly when the system must deal with unpredictable events and conditions that could not be foreseen or even modeled during testing.

One approach to CAIS assurance is to build a non-critical prototype (or digital twin) that can safely model the behavior and intended purpose of the CAIS system, and to build the test suite using the model prototype. The prototype must be correct – as the degree to which it is correct is a system trust consideration [1]. The proxy would be what can be called "metaphorically equivalent" to a real CAIS system. For example, an autonomous vehicle/robot used in a fire emergency could be modeled by a robot garbage collector proxy. The proxy could be used test mis-navigation errors or identification errors during extreme heat.

To build these types of prototypes, the underlying characteristics of any CAIS need to be understood and modeled rigorously. This understanding also informs overall system assurance because if the characteristics of the CAIS are understood, then they can be verified and validated appropriately.

In this work we address the problem of assurance for CAIS systems by developing a focused taxonomy of CAIS characteristics and operations. This taxonomy enables the development of the non-critical proxy (i.e., stand in) equivalent by matching characteristics and operations. Ultimately, this approach can lead to a set of CAIS proxies, which in turn informs the selection of the "right" proxy to help find errors that could manifest in the deployed CAIS..

II. Background: Autonomous Systems

Before a system is analyzed for a functional equivalent proxy, the level of autonomy and criticality of the system should be assessed. These aspects are interrelated as many critical systems have a great deal of autonomy. For example, systems that interpret data to generate a solution such as a medical diagnostic system or autonomous vehicles can be deemed critical given the interaction with human life. And by definition, occupied autonomous vehicles have a high level of autonomy and criticality given the possible consequences of a vehicle crash.

In September 2021, the American Medical Association (https://www.ama-assn.org/) added guidance to classify AI applications: assistive (machine detects relevant data without analysis or generative conclusions), augmentative (machine analyzes and/or quantifies data), or autonomous (automatically interprets data and independently generates clinically relevant conclusions).

The taxonomy further classifies the level of AI in a particular machine: level 1 (draws contestable conclusions but requires a physician to implement, ex. imaging), level 2 (draws conclusions in a clinically meaningful way and initiates diagnosis but can be overridden, ex. continuous glucose monitoring), level 3 (initiates disease management, ex. retinal imaging). One could compare a CAIS to a level three autonomous medical system shown in Table 1:

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TABLE I. AMERICAN MEDICAL ASSOCIATION AI TAXONOMY FOR MEDICAL SERVICES AND PROCEDURES [2]

Taxonomy Component	Assistive Level 1	Augmentative Level 2	Autonomous Level 3
Primary Objective	Detects clinically relevant data	Analyzes// quantifies data in a meaningful way	Interprets data and generates conclusions
Independent diagnosis and/or decision management?	No	No	Yes
Data Analysis	No	Yes	Yes
Requires interpretation of professional	Yes	Yes	No

Another option to assess the criticality of a system was provided by the European Union (EU). In an effort to manage AI risk to the public, the EU which is an economic and political union between 27 European countries, proposed AI regulation rules to facilitate global AI regulation. The regulations divided AI systems into three categories shown in Table 2: unacceptable-risk, high-risk, and limited/minimal risk [3]. One could compare a CAIS to the high-risk and unacceptable-risk categories shown in Table 2.

TABLE II. EU AI RISK CLASSIFICATION SYSTEM

Limited- and minimal-risk	High-risk	Unacceptable-Risk
• AI Chatbots • AI-enabled video and computer games • Spam filters • Inventory-management systems • Customer- and market-segmentation systems • Most other AI systems	• Systems that evaluate consumer creditworthiness • Recruiting or employee-management systems • Systems utilizing biometric identification in nonpublic spaces • Safety-critical systems (systems that could put citizen health at risk) • Any systems use in the administration of justice	• Subliminal, manipulative, or exploitative techniques causing harm • Real-time, remote biometric identification systems used in public spaces for law enforcement • All forms of social scoring (evaluates a person's trustworthiness based on social behavior or predicted personality traits).

Once a system is classified as a CAIS, the goal is to determine the metaphorically equivalent system (equivalence of application/PROXY). The proxy should have the functional equivalence of the CAIS to allow a safe way to test the critical system. For example, a robot vacuum may have significant operational and implementation similarities to act as a testing proxy for some use cases of an autonomous vehicle.

The performance and effective testing of the proxy highly depends on choosing the correct set of features. To determine the functional equivalence a feature extraction process using the taxonomy described in the next section could be utilized to identify attributes of the system.

III. CAIS TAXONOMY

The taxonomy described in this section is used determine the characteristics and classify the CAIS and to map the characteristics of the critical AI system to its testing proxy (non-critical prototype/digital twin). In other words, this taxonomy will help to assess functional equivalence of the testing proxy.

The CAIS taxonomy includes five dimensions: operational environment, AI application purpose, operational characteristics, AI/ML technologies, and AI/ML techniques.

A. Physical Operational Environment:

Operational environments (OE) are described by the US Department of Defense as, "a composite of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander," [4]. Generally operational environments include air, space, and subsurface (maritime, oceanography, hydrology terrain). When considering these environments both the positive and negative effects on the system should be evaluated. Note: "Cyberspace" or an "informational environment" could be considered here, given data and information travels to and from the physical world to the cyber world, this environment is addressed in the other dimensions.

B. AI Application Purpose:

The application purpose dimension of the CAIS taxonomy utilizes an adaptation of the taxonomy shown in Table 3. The domains listed are a result of a qualitative review of AI literature from 1955 to 2019 by the European Commission [5]. Their taxonomy is a collection of keywords, with some overlap, that characterize the domains of AI research.

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TABLE III. EUROPEAN COMMISSION AI APPLICATION TAXONOMY

AI Domain	Subdomain	Example Key Words within the subdomains
Reasoning	Knowledge representation	Case-based reasoning, inductive programming, expert system etc.
	Automated reasoning	
	Common sense reasoning	
Planning	Planning and Scheduling	Bayesian optimization, constraint satisfaction, evolutionary algorithm etc.
	Searching	
	Optimization	
Learning	learning	Deep learning, data mining, etc.
Communication	Natural language processing	Chatbot, computational linguistics, conversation model etc.
Perception	Computer vision	Action recognition, face recognition, image processing etc.
	Audio processing	Computational auditory scene, music information retrieval, sound description etc.
Integration and Interaction	Multi-agent systems	q-learning, intelligent agent, game theory, etc.
	Robotics and Automations	Cognitive system, control theory, human-ai interaction etc.
	Connected and Automated vehicles	Autonomous vehicles
Services	AI Services	Data analytics, decision support, big data, virtual reality etc.

C. Operational Characteristics:

Operational characteristics represent potential behaviors and effects of the system. There are many possible ways to organize a set of operational characteristics and these will need to be standardized over time. But a simplified set of operational characteristics and associated metrics/measures are given below.

O1. Moving/stationary [no=0/yes=1].

O2. Mission: navigation, target acquisition, target attack, gathering something, delivering something/payload (gas, water, packages) [can be >1 of these; b1b2b3b4b5 where bi=1 if the domain applies]

O3. Financial consequences [on a scale of 0-9 where 0 represents no financial consequences and 9 represents catastrophic financial consequences]

O4. Social consequences [on a scale of 0-9 where 0 represents no social consequences and 9 represents catastrophic social consequences (e.g., privacy, elections, compliance/law)

O5. Human Risk [on a scale of 0-9 where 0 represents no human risk and 9 represents catastrophic human risk (e.g., to the operator, user, passenger)

D. AI/ML development algorithms

This dimension is mapped to both the reasoning and planning domains in Table 2 which refers to all types of reasoning as well as planning/searching/scheduling/optimization.

AI and ML, terminology that is often used interchangeably, are software system algorithms that use data to predict outcomes. The initial algorithm chosen depends on the task at hand. There are many to choose from and they all have their benefits. To converge on a more accurate proxy, the proxy should use/match the same algorithm as the system being mimicked. Here are example algorithms with an example of its wide usage [6]:

- Naïve Bayes (ex. text classification)
- Support Vector Machine (ex. pattern classification problems)
- Linear Regression (ex. business sales forecasting)
- Logistic Regression (ex. weather forecasting)
- K-Nearest-Neighbor (KNN) (ex. market segmentation, computer vision, astronomy etc.)
- Decision Tree (ex. used to understand and interpret data)
- Random Forest (ex. used for ensemble learning)
- Cart (another type of decision tree)
- Apriori Machine Learning Algorithm (ex. sorting large amounts of data)
- Principal Component Analysis (ex. image processing)
- CatBoost (ex. can be combined with deep learning frameworks and numerous data types)
- Iterative Dichotomiser 3 (used for machine learning and linguistic communication)
- Hierarchical Clustering (used to illustrate data)
- Back Propagation (used for classification and regression)
- AdaBoost (used with other ML algorithms to increase performance)
- Deep Learning (used for deep learning in text classification)
- Gradient Boosting Algorithm (used to develop predictive models)
- Hopfield Network (a type of recurrent artificial neural network)
- C4.5 (a type of decision tree)

E. AI/ML development techniques:

It is important that the techniques used to develop the CAIS match the techniques used in the proxy implementation. The techniques refer to the programming languages (Javascript, Java, C++, etc.), development environments, and software development process (agile, waterfall etc.)

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It is important to note that "matching" these 5 operational characteristics is vital for each proxy system accuracy. Every environment, technique, characteristic, algorithm, and purpose of the proxy system needs to be clearly and thoroughly defined to represent each CAIS characteristic.

To demonstrate, when building a proxy for an autonomous system, it is important that the CAIS system is by definition an autonomous/unmanned system – and further, what type of unmanned system is it. For example, in the National Institute of Standards and Technology (NIST) SP- 1011-I-2-0 (2007), the Department of Defense definition of unmanned (autonomous) vehicles was used to define levels of unmanned systems: "no human operator aboard the principal components, which acts in the physical world to accomplish assigned tasks. It may be mobile or stationary. It can include any and all associated supporting components such as operated control units (OCU)s." They provided examples: unmanned ground vehicles (UGV), unmanned aerial vehicles/systems (UAV/UAS), unmanned maritime vehicles (UMV) – unmanned underwater vehicles (UUV) or unmanned water service borne vehicles (USV) – unattended munitions (UM), and unattended ground sensors (UGS). They also specified that missiles, rockets, and their submunitions, and artillery are not considered the principal components of unmanned systems. Then it needs to be determined if the system is full or semi-autonomous where semi is defined as an unmanned system that is capable of autonomous operation between human interactions [7].

IV. CAIS Taxonomy Template

The taxonomy's five dimensions are used to match the characteristics of CAIS to determine a suitable proxy system. The template shown in Table 4 assists in determining a system's distinguishing features. In Table 5, each dimension is weighted to understand the most vital dimensions (bold and underlined text) of the CAIS proxy. For example, the development technique may have less impact on a proxy decision than the operational characteristic, therefore, prioritizing those requirements in the proxy design and implementation. Tables 4 and 5 demonstrate the use of the CAIS taxonomy with a DNA matching algorithm- a critical system given financial consequences due to lawsuits in the event of failure, eroding trust and possible loss of life if DNA therapy is incorrect.

TABLE IV. Example CAIS Template use

	Phy. Op. Envmt.	**AI App. Purpose**	**Operational Charac.**	**Dev. Algorithm**	**Dev. Tech.**
DNA Matching	Online	Reasoning, learning, planning, services	O1:0; 02: 0; O3: 5; O4: 9; 05: 9	KMP Algorithm	Java

TABLE V. Taxonomy Dimension Weight

	Noncritical	*Somewhat Critical*	*Critical*
Physical Operational *Environment*	x		
Application: AI Purpose		x	
Operational Characteristics			**X**
Development Algorithms			**X**
Development Techniques		x	

V. Proxy Testing

To gain the highest level of confidence in the proxy it needs to be tested. Proxy testing could be compared to the re-certification that occurs in critical avionics software. For example, when a software upgrade occurs in avionics software, the question becomes whether the entire system needs to be recertified or just the parts that can be proven to be statically "touched/reachable" by the modifications.

"Similarity" metrics could be used to determine reused functionality (code or algorithms) between the proxy and the CAIS that are deemed "similar." In other words, If the proxy system actually uses a lot of the same code as the CAIS system this would give one better confidence in the testing. Imagine if the only difference between the robot weed killer and the robot soldier was a few modules of code (e.g., the difference between identifying plants and people – same algorithm but different parameters) and the delivery of the weapon (weed killer versus bullet).

VI. Conclusion

A five-dimensional taxonomy to design and develop CAIS proxies was presented. Use of this taxonomy provides a tool to critically analyze the dimensions of a CAIS in order to build an equivalent proxy for testing. This taxonomy enables the development of the non-critical proxy (i.e., stand in) equivalent by matching physical operational environments, application purpose, operational characteristics, development algorithms, and development techniques. The weighting system provides additional analysis to determine the most critical dimensions of the proxy. Further analysis may determine any mappings between the taxonomy and dimension weighting. Ultimately the CAIS taxonomy and analysis technique presented in this paper will lead to a set of CAIS proxies that can provide the vital information to select the appropriate proxy where the proxy can be used for critical error testing.

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